

Lecture 2 - Induction and Cardinality

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- A natural number n , is a whole number which is greater than or equal to 1.
 - Denoted $n \in \mathbb{N}$.
- The principle of mathematical induction (or just induction) is used to prove statements about all natural numbers.
 - Induction can be given in two formulations:
 1. The set formulation of the principle of mathematical induction.
 2. The statement formulation of the principle of mathematical induction.
 - Note that induction only works on sets with the well-ordering property.
- A statement $P(n)$ about a natural number n is an expression which can either be true or false.
 - The statement formulation of induction says that, for some hypothesis about natural numbers $P(n)$, that if some base case $P(b)$ is true, and that $\forall k \in \mathbb{N}, k \geq b, P(k) \implies P(k+1)$, then $P(n)$ must be true for all $\forall n \in \mathbb{N}, n \geq b$.

1 Prove that the statement $P(n) : 1 + 2 + 3 \dots + n = \frac{(n)(n+1)}{2}$ is true $\forall n \in \mathbb{N}$ by induction.

Proof. We do this in 3 steps:

(1) Prove the base case (Show $P(1)$ is true).

LHS:

$$= 1$$

$$= 1$$

RHS:

$$= 1 \frac{(1)(1+1)}{1}$$

$$= 1$$

Therefore, $P(1)$ is true.

(2) Formulate our induction hypothesis (State $P(k)$).

Here, we assume that our statement holds true for some $k \in \mathbb{N}$. Therefore, our induction hypothesis can be formulated as:

$$P(k) : 1 + 2 + \dots + k = \frac{k(k+1)}{2}$$

$$(3) \text{ Prove } P(k) \implies P(k+1)$$

LHS:

$$\begin{aligned} &= 1 + 2 + \dots + k + k + 1 \\ &= \frac{(k)(k+1)}{2} + k + 1 \\ &= \frac{(k)(k+1)}{2} + \frac{2k+2}{2} \\ &= \frac{k^2 + k + 2k + 2}{2} \\ &= \frac{k^2 + 3k + 2}{2} \\ &= \frac{(k+1)(k+2)}{2} \end{aligned}$$

RHS

$$= \frac{(k+1)(k+2)}{2}$$

Therefore, since the $LHS = RHS$, we know that $P(k) \implies P(k+1)$. Since $P(1)$ is true, then $P(1) \implies P(2) \implies \dots P(n)$ and thus the statement $P(n)$ holds for all $n \in \mathbb{N}$ by principle of mathematical induction. □

2 Prove that $1 + r + r^2 + \dots + r^n = \frac{1-r^{n+1}}{1-r}, r \in \mathbb{R}, n \in \mathbb{N}, r \neq 1$

Proof. We formulate our proof as before, first by fixing $r \neq 1$ and defining our statement as $P(n) : 1 + r + r^2 + \dots + r^n = \frac{1-r^{n+1}}{1-r}$.

$$(1) \text{ Prove the base case (Show } P(1) \text{ is true).}$$

LHS:

$$\begin{aligned} &1 + \dots + r^n \\ &= 1 + r \end{aligned}$$

RHS:

$$\begin{aligned} &\frac{1 - r^{n+1}}{1 - r} \\ &= \frac{1 - r^{1+1}}{1 - r} \\ &= \frac{1 - r^2}{1 - r} \\ &= \frac{(1+r)(1-r)}{1-r} \\ &= 1 + r \end{aligned}$$

Therefore, $P(1)$ is true.

(2) Formulate our induction hypothesis (State $P(k)$).

Here, we assume that our statement holds true for some $k \in \mathbb{N}$. Therefore, our induction hypothesis can be formulated as:

$$P(k) : 1 + r + \dots + r^k = \frac{1 - r^{k+1}}{1 - r}$$

(3) Prove $P(k) \implies P(k+1)$

Here, as opposed to trying to show the $LHS = RHS$, I'll instead just show that $1 + r + \dots + r^k + r^{k+1} = \frac{1 - r^{k+2}}{1 - r}$ using our induction hypothesis.

$$\begin{aligned} & 1 + r + \dots + r^k + r^{k+1} \\ &= \frac{1 - r^{k+1}}{1 - r} + r^{k+1} \\ &= \frac{1 - r^{k+1}}{1 - r} + \frac{(1 - r)r^{k+1}}{1 - r} \\ &= \frac{1 - r^{k+1} + r^{k+1} - r^{k+2}}{1 - r} \end{aligned}$$

Therefore, $P(k) \implies P(k+1)$. Since $P(1)$ is true, then $P(1) \implies P(2) \implies \dots P(n)$ and thus the statement $P(n)$ holds for all $n \in \mathbb{N}$ by principle of mathematical induction. $\square$

- The principle of strong mathematical induction is another technique used to prove statements about the natural numbers.
 - In the principle of strong induction, rather than assuming that $P(k)$ holds for some $k \in \mathbb{N}$, we instead assume $P(1) \wedge P(2) \wedge \dots \wedge P(k)$ and attempt to prove that $P(1) \wedge P(2) \wedge \dots \wedge P(k) \implies P(k+1)$.

3 Suppose (x_n) is a sequence, $x_n \in \mathbb{Q} \forall n \in \mathbb{N}$ and (x_n) is defined as $x_1 = 1, x_2 = 2$, and $x_n = \frac{x_{n-1} + x_{n-2}}{2}$. Prove that $1 \leq x_n \leq 2 \forall n \in \mathbb{N}$ by strong induction.

Let $P(n) : 1 \leq x_n \leq 2$.

(1) Prove the base case.

Since we are dealing with strong mathematical induction, as you will see in our final step of the proof, we need to be sure that both $P(1)$ and $P(2)$ hold true. Thankfully, both statements are trivially true.

$$\begin{aligned} x_1 &= 1 \\ x_2 &= 2 \end{aligned}$$

Therefore, since $1 \leq x_1 \leq 2$ and $1 \leq x_2 \leq 2$, $P(1)$ and $P(2)$ both hold.

(2) Formulate our induction hypothesis (State $P(k)$).

Since we are attempting to prove our statement by strong induction, we assume that there exists some $k \in \mathbb{N}$ such that $P(1) \wedge P(2) \wedge \dots \wedge P(k)$ is true. Therefore, our induction hypothesis is given by:

$$1 \leq x_1 \leq 2 \wedge 1 \leq x_2 \leq 2 \wedge \dots \wedge 1 \leq x_k \leq 2$$

$$(3) \text{ Prove } P(1) \wedge P(2) \wedge \dots \wedge P(k) \implies P(k+1)$$

$P(k+1)$ is the statement $1 \leq \frac{x_k + x_{k-1}}{2} \leq 2$. By the definition of (x_n) , we know that $x_{k+1} = \frac{x_k + x_{k-1}}{2}$. Now since we are assuming $P(k-1)$ and $P(k)$, we assume that $1 \leq x_{k-1} \leq 2$ and $1 \leq x_k \leq 2$, and therefore:

$$\begin{aligned} x_{k+1} &= \frac{x_k + x_{k-1}}{2} \implies \\ \frac{1+1}{2} &\leq \frac{x_k + x_{k-1}}{2} \leq \frac{2+2}{2} \\ \frac{2}{2} &\leq \frac{x_k + x_{k-1}}{2} \leq \frac{4}{2} \\ 1 &\leq \frac{x_k + x_{k-1}}{2} \leq 2 \end{aligned}$$

Therefore, $P(1) \wedge P(2) \wedge \dots \wedge P(k) \implies P(k+1)$. Thus, since $P(1)$ and $P(2)$ holds, then by principle of strong mathematical induction, $P(n)$ holds $\forall n \in \mathbb{N}$.

- A function $f : A \rightarrow B$ denotes that a function has domain A and co-domain B .
 - f is a *surjective* (onto) function if $\forall b \in B, \exists a \in A$ such that $f(a) = b$.
 - f is an *injective* (1 to 1) function if $\forall a_1, a_2 \in A, a_1 \neq a_2, f(a_1) \neq f(a_2)$.
 - f is *bijective* if it is both injective and surjective.
- The set $\mathbb{N}_n$ denotes the first n natural numbers.
- The cardinality of a set represents the size of a set.
- A set can be either *finite* or *infinite*.
 - A set S is finite if it is either the empty set $\emptyset$ or if there exists a bijection between S and $\mathbb{N}_n$ for some unique $n \in \mathbb{N}$.
 - * S has a cardinality of 0 if $S = \emptyset$ or n for the bijection between S and $\mathbb{N}_n$.
 - * The cardinality of a finite set is denoted $\#S$.
 - A set S is infinite if there exists no bijection between S and $\mathbb{N}_n \forall n \in \mathbb{N}$.
 - * An infinite set can be either *denumerable* or *uncountable*.
 - A set is denumerable if there exists a bijection between $\mathbb{N}$ and S .
 - A set is uncountable if there exists no bijection between $\mathbb{N}$ and S .
 - * A set is *Countable* if it is either finite or denumerable.

We can visually represent the different kinds of sets with the following Venn diagram:

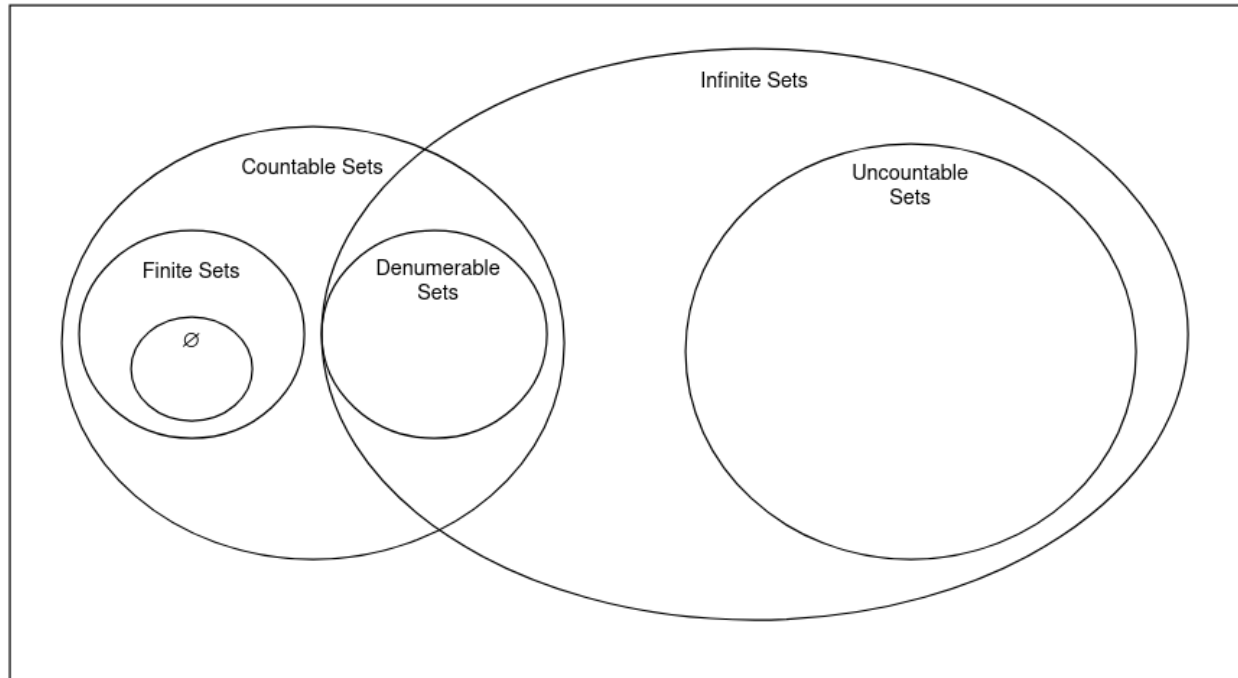


Figure 1: Venn Diagram representing the different kinds of sets.